

Fall 2025 Wind Tunnel Test

RAHUL GUPTA

Overview

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Project Context & Test Objectives

Context

- Conducted as a final project for Low-Speed Lab (UT ASE 120K) which allowed us to choose our model and testing setup in the ASE Subsonic Wind Tunnel
- Used the opportunity to run a wind tunnel test for Longhorn Racing (FSAE) Electric Aerodynamics to correlate CFD and real-world results

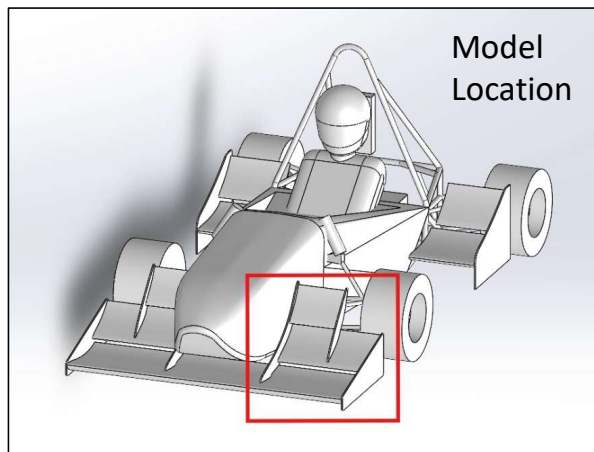
Objective

- Correlate CFD data with wind tunnel data
 - Correlate CFD stall and model stall using a -10° to 30° angle-of-attack sweep
 - Correlate smoke wand data with CFD visuals to roughly validate vorticity modeling
 - Correlate pistol grip lift/drag measured forces with expected forces through RPM and angle-of-attack sweeps

Model & Manufacturing

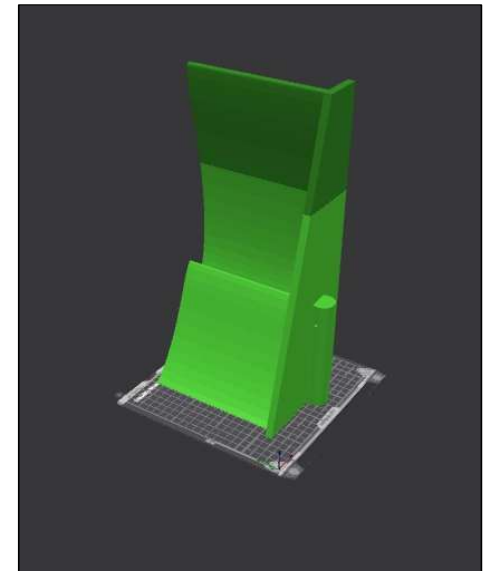
Model Selection

- Used the outboard of nosecone section of the 25-26 front wing
- Cut off the third element and shortened the wingspan
- 68% scale model, blockage ratio < 5%



Manufacturing

- Needed to be rapid because of 2-day turnaround time (for class project)
- 3D printed in two sections using PLA, combined using a press fit, adhesive, and tape



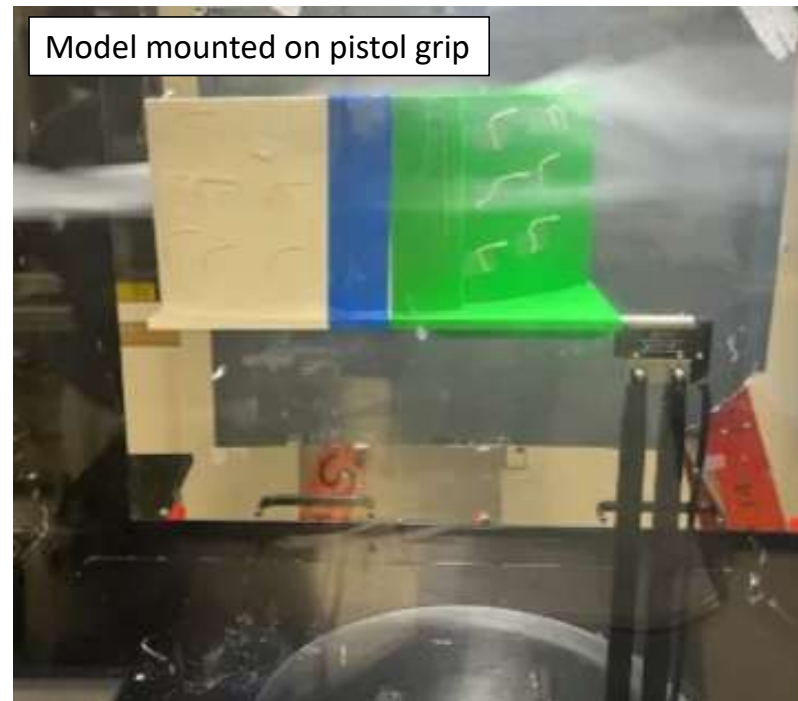
Mounting & Measurement

Pistol Grip Force and Moment Balance

- Used measurements for the pistol grip existing in the ASE wind tunnel to create the boom on the endplate
- Mounted with span outward from endplate in +z direction, leading edge facing +x direction
- Pistol grip mount allowed for angle-of-attack change through rotation about the z axis

Qualitative Measurement

- Smoke wand upstream
- Tufts taped to model surface



Quantitative Results

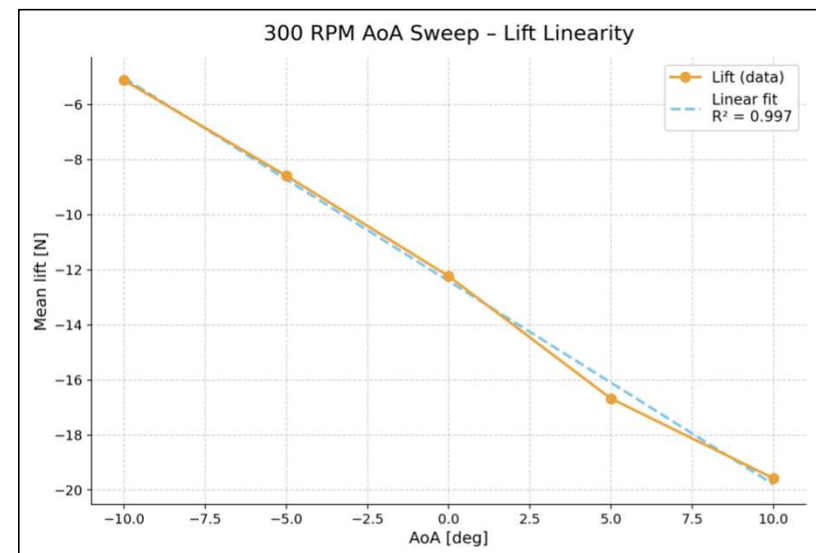
Achieved

- Constant C_l , C_d across RPMs
- Lift and Drag linearity with angle of attack

Failed to achieve (see slide 8)

- Data past 400 RPM
- Useful data at high angle-of-attack

Case	V (m/s)	$C_L \cdot S$ (m ²)	$C_D \cdot S$ (m ²)	C_L	C_D
300 RPM	9.07	-0.0580	0.0311	-2.36	1.26
400 RPM	12.25	-0.0597	0.0310	-2.43	1.26



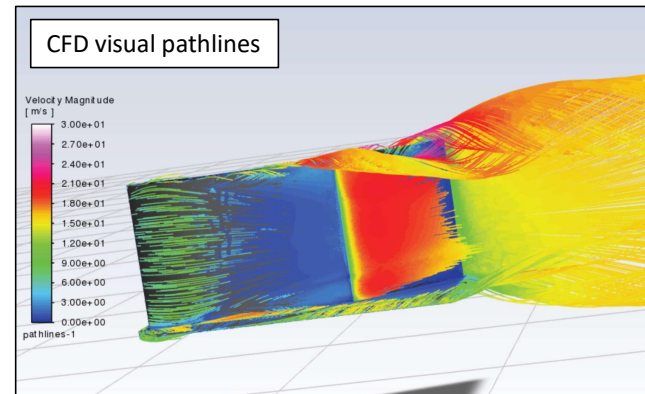
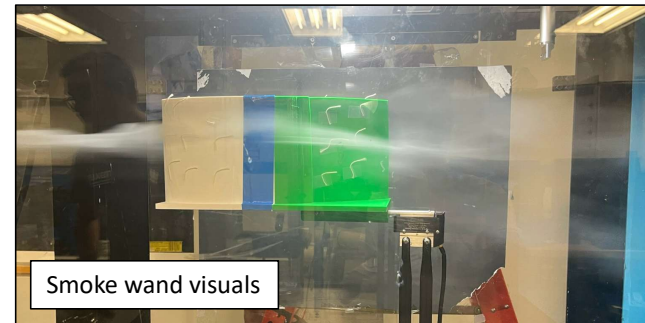
Qualitative Results

Evidence from limited testing

- Proof of similar flow structure in CFD and in wind tunnel
- Shown with smoke wand and oscillatory tuft behavior

Limitations

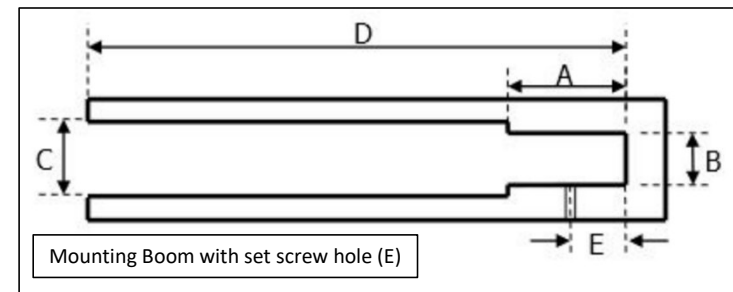
- Similar qualitative structures are not very conclusive for correlation between CFD and real-world airflow
- Aimed to use qualitative data to find stall point, was ineffective at doing that (see slide 8)



Future Test Improvements

Rotation about Pistol Grip

- Set screws couldn't prevent rotation about x-axis at higher speeds or angle-of-attack's
- Possible Solutions:
 - Aerodynamic Boom at Midplane for mounting the grip
 - Keep cascade horizontal with span along the y-axis
 - Keep model symmetric for force/moment balance

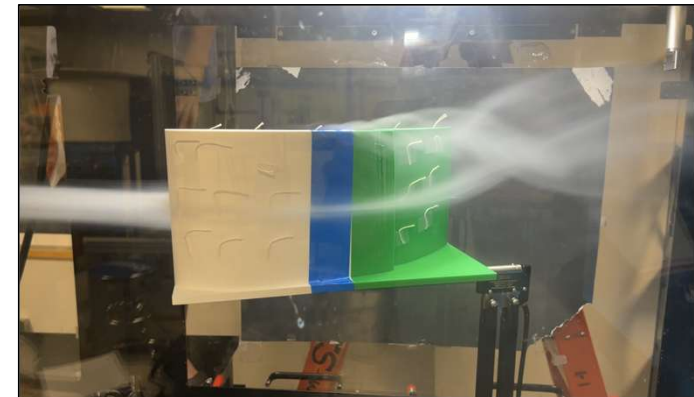


Failure to Capture High AoA Separation

- Lack of endplates caused high vorticity generation, which kept flow attached on bottom surface, preventing noticeable separation
- Possible Solutions:
 - Use two endplates

Future test goal

- Be able to test full range of RPMs and AoAs



Future Test Improvements

Model Smoothness

- Used duct tape and glue to assemble
 - Properly dimensioned/tolerance press fit sockets + extrusions
- Trailing edge thinning and fraying
 - Fillet/chamfer and model those in CFD
 - Change/experiment with print settings
 - Use automotive filler and rtv to fill out and seal the trailing edge
- Print lines
 - Sand down, or acetone wipe to smooth (careful not to warp profile)

Possible Structural Vibrations

- Oscillation in the data indicates mounting issues
 - Magnitude of oscillation indicates a structural issue rather than an oscillatory aerodynamic effect

